

❑ 12 What is/are statistics?

Statistics is concerned with the acquisition, analyses, interpretation, and presentation of data.

What do we mean by data? Numbers yes, but not just numbers!

Example

Consider an investigation to explore the factors that effect a crop's growth on farms in a region with the following information collected.

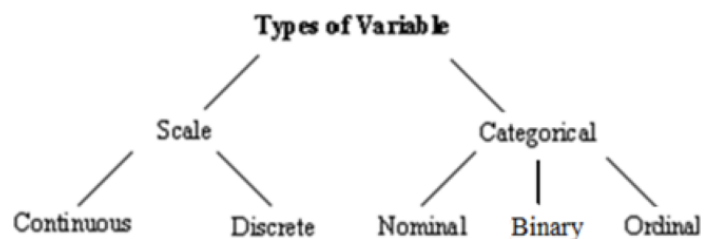
- average rainfall (mm)
- total sunshine (days)
- fertilizer used (none, Brand A, Brand B, Brand C)
- soil type (sandy, clay, peat)
- farmer / farm manager has a higher-level qualification in agricultural / horticulture (yes, no)
- crop yield (tonne)
- crop quality (poor, fair, good, excellent)

Each of these factors gives rise to a variable (e.g., sunshine, soil type etc.)

The values taken by a variable on any one farm is the data. (E.g., 20 days sunshine, soil type sandy etc.)

• Classifications of variables

N.B. The distinction between scale and categorical (sometimes called factors) is the most important.



Scale?

Data is a numerical value that represents an amount or quantity. I.e., it can be located on a number scale but is *not* used as a label.

In our example, rainfall, sunshine and mass are scale variables.

Also, quantities such as time, weight, length, volume, density, percent, counts etc.

But for example, diabetes type I or II or FIFA qualifier group 7 etc. are **not** scale variables.

Scale (discrete)?

E.g., number of slugs on a plant may be 0, 1, 2, but not 1½!

Scale (continuous)?

E.g., total mm of rainfall in a day is not similarly limited.

1 mm, 2 mm but also 1.5, 1.54 etc.

Notes

1. Discrete quantities are frequently, but not always integer values – for example, counts. There are some exceptions – for example, shoe sizes (8, 8½, 9 etc.)
2. Some quantities are technically discrete, but it will be convenient to regard them as continuous. (It could be argued that all data is discrete –e.g., our technique for measuring rainfall may not permit us to obtain a measured value between 1.5 and 1.6.)

For the purposes of this module, if we reason (subjectively) that there is no meaningful difference between two successive values for a discrete quantity, we will regard it as continuous.

For example, consider a restaurant charge for €53.77. The next point on the (discrete) currency scale would be €53.78. But €53.78 is not meaningfully different from €53.77 so in this context, we would regard restaurant charge as being a continuous scale variable.

But for two competing office supply firms charging 2c and 3c per sheet of A4 paper, we would likely regard paper charge as being a discrete scale variable.

A doctor assessing an adult patient may see no meaningful difference between ages of say 43, 44 or 45. But a paediatrician clearly would distinguish between children aged 2, 3 or 4.

Categorical (nominal)?

Data may be classified into one of a number of unordered categories. The distinct categories are the levels of the variable. Above, soil type (3 levels) and fertiliser used (4 levels, not 3 as no fertilizer counts as a level).

Other examples: crop variety, genotype, food group, bacteria type, colour, brand, religion, nationality, gender, occupation, college course, production line (1, 2 or 3), raw material source (A or B) etc.

Categorical (ordinal)

Ordinal variables are similar to nominal, but the levels have a logical ordering.

In our example: crop quality. Excellent > Good > Fair > Poor

Other examples include:

Education level	2nd level < B.Sc. < M.Sc. < Ph. D
Army ranks	Private < Sergeant < Lieutenant < Colonel
Manufacturing firm employees	Operator < Line supervisor < Plant manager < CEO
Likert scales	strongly disagree < disagree < no opinion < agree < strongly agree
Last birthday age ranges	0 - 20 < 21 - 30 < 31 - 40 , ...

Note that with ordinal variables we cannot quantify the difference between levels.

E.g., we cannot say with any authority that “strongly disagree” is lower than “disagree” by a greater amount than “disagree” is lower than “no opinion”.

● **Acquiring data**

Data used in any analysis might be primary (collected by us) or secondary (collected by someone else). Secondary data is convenient (acquiring primary data can be expensive) but it may not be possible to verify that the data was acquired in an appropriate way.

Primary data is typically acquired either by sampling (e.g., political opinion polls, survey of fish stocks in waterways etc.) or by generation in an experiment (e.g., industrial processes experiments, scientific or engineering investigations etc.)

When data is collected by sampling, care must be taken to ensure that the sample is representative of the population from which it is taken. Similarly, when data is generated from designed experiments there are pitfalls that must be avoided if the integrity of the data is to be sound.

Example – customer survey data

An online trading site sells goods that may be classified as electronics, jewellery or fragrances. After every purchase, customers are invited to complete a survey reporting such information as their age, sex, type and cost of good purchased, how they paid (credit card, PayPal etc.) their satisfaction with the experience using the web site etc. The objective here might be to identify general patterns in customer’s behaviour. (E.g., do males spend more than females? do customers who buy electronic goods tend to be younger? etc.)

Example – process experiment

A distributor of fuel products is developing a new, peat-based fire log product and is devising an experiment to explore a number of factors that may impact on the longevity (burn time) of the log. The factors are *supplier* (from what source – A, B or C was the peat used in the logs obtained), *accelerant* (whether or not the log was coated with a fire accelerant), *compression* (the force used to compress the peat into a log) and *boreholes* (the number of bore holes (to aid airflow) drilled into the logs). The objective of the study might be to identify settings (levels) for the various factors that would maximise longevity.

Exercise

Identify the types of the variables in the process experiment example.

• Experimental v observational variables

The process experiment example requires that the data be generated. Note that it is within the gift of the investigator to choose (or experiment with) the levels of the factors. She can decide to consider one run and elect to use peat supplied from B, to not use an accelerant, to use a compression force of 5.6 kN and to have two boreholes in the log. Thus, all four factors give rise to experimental variables

The customer satisfaction survey requires that a sample be taken from a population (i.e., the population of all customers of this web site) but the surveyor does not dictate what the levels or values of the variables should be – they simply observe and so the variables generated are observational variables.

Note that in the crop growth example all of the variables with the possible exception of *fertilizer* are observational. (Those carrying out the study could direct what fertilizer be used on each farm – but they can only observe the amount of rainfall, sunshine, the soil type etc.)

Frequently the objective in a statistical study is to make a claim for causation (e.g., higher intakes of fatty food lead to higher cholesterol, the higher the operational speed of an engine, the lower the fuel efficiency etc.) It is particularly challenging to make a case for causation with observational variables.

• Structuring raw data

To facilitate the analyses of raw data, however it is sourced, the data is usually arranged in a table with variables in columns and records (e.g., subjects/respondents, experimental runs) in rows as the following example illustrates.

The Titanic data set – an example of secondary data

The Titanic was a passenger liner that sank in 1912 with over 2,200 people on board.

Five variable measures on the passengers are recorded and are data for the first few people are tabulated opposite.

age	sex	embarked	class	survived
22	female	southampton	3rd	no
36	male	cherbourg	1st	yes
.	male	NA	NA	no
44	male	queenstown	1st	no

Note the structure of the data with variables in rows and subjects or records or cases in rows. This is typical of how data should be arranged for analysis in most statistical software packages. Note too that some data is NA (not available/applicable). Such a structure is sometimes described as stacked.

Exercise Can you identify the type of each variable?

Data may alternatively be arranged using an unstacked format. The following example shows data for exam scores attained by 3 males and 4 females.

Sex	Score
Male	62
Male	78
Female	45
Female	61
Male	57
Female	49
Female	83

Stacked format

Male	Female
62	45
78	61
57	49
	83

Unstacked format

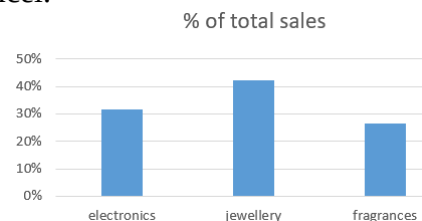
• Descriptive v Inferential statistics

Descriptive or summary statistics focus only on the data collected and use numbers (called statistics), charts or tables to extract and convey information in the data.

For example, in the crop yield experiment the investigator might calculate that the average crop yield across all 8 farms was 4.3 tonne per acre and that the 4th farm has the maximum yield with 7.1 tonne per acre (the average and the maximum are examples of statistics)

Or an analysis of the customer survey data might compile a percentage frequency table showing the popularity of the three different product types (below left) and from this construct a plot of the data - called a bar chart (below right). Both are constructed using Microsoft Excel.

Category	<i>electronics</i>	<i>jewellery</i>	<i>frangrances</i>
% of total sales	32%	42%	26%



With inferential statistics we seek to infer or generalise about an entire population from a sample.

E.g., the researcher in the crop yield experiment may forecast total output of this crop from the region by inferring that output observed in the sample will be similar in the population – i.e., 4.3 tonne on average.

Or in the process experiment, the engineer might infer that the same log burn time observed in the experiments for particular factor level combinations would also be observed in an actual production run.